PCB Design Report

SparkFun OpenLog Artemis Data Logger (Round 2, Problem 1)

Submitted By-

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EDA Tool: **EasyEDA**

Contest: iTest Bangladesh PCB Design Contest 2026, Round Two

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1. Introduction

This report describes the design of a compact data-logging PCB based on the SparkFun OpenLog Artemis. The board has an Artemis module, an ICM-20948 IMU, a microSD socket, a USB-C interface with a CH340E serial bridge, a LiPo charger with power management, a Qwiic connector, and debug and status circuits. Sensitive sensor circuits are kept away from noisy power and USB sections.

2. Component List (BOM)

Table 1 lists all components with designator, value, package and source. Parts from the EasyEDA library are marked "Library"; parts built from datasheet dimensions are marked "Custom".

Designator	Component	Value	Package	Source
U8	Artemis module (ARTEMIS_MODULE)	-	ARTEMIS_FP [module SMD]	[Library / Custom]
U13	9DoF IMU ICM-20948	-	QFN-24, 3x3 mm	Library
U10	USB-to-serial CH340E	-	MSOP-10	Library
U2	LiPo charger MCP73831	-	SOT-23-5	Library
U3, U4	3.3V LDO AP2112K-3.3	3.3 V	SOT-23-5	Library
U14	1.8V LDO (SP6214 equivalent)	1.8 V	SOT-23-5	Library
U1	3.0V supervisor	3.0 V	SOT-23-3	Library
Q2	P-MOSFET (microSD power switch)	20V / 4.2A	SOT-23	Library
Q1, Q3, Q7, Q9, Q10	N-MOSFET (level shifting)	20V / 0.2A	SOT-23-3	Library
D1, D3	Schottky diode	3A / 10V	SMB	Library
D6	Diode	1A / 23V	[package]	Library
Y3	Crystal 32.768 kHz	32.768 kHz	SMD 2-pad	Library
J5	USB Type-C connector	-	16-pin SMD	Library
J6	microSD socket	-	Push-push, R/A	Library
J11	Qwiic connector (JST 4-pin, 1 mm)	-	SM04B-SRSS-TB	Library
J2	LiPo connector (JST 2 mm)	-	S2B-PH-SM4-TB	Library
J14	Cortex JTAG/SWD header	2x5, 1.27 mm	SMD 2x5	Library
J1, J3, J4, J7	Header pins	2.54 mm	1x2, 1x6, 1x1	Library
S3	Reset tactile switch	-	4-pin SMD	Library
BT1	RTC battery ML414H	-	ML414H holder pads	Library
JP2, VE1	Jumpers (MEAS, SMT jumper)	-	[solder jumper]	[Library / Custom]
R1-R8, R22-R29, R40, R41, R44-R49	Resistors (R49 = DNP)	[values]	0402	Library
C1-C5, C7, C29-C33, C40-C44, C46, C51-C54	Capacitors	[values]	0402	Library
[LED designators]	Status LEDs (red, green, blue)	-	0805	Library
GND, QWIIC_3.3V, SCL, SDA, TP1	Test points	-	Pad D1.5 mm	Library

Table 1: Bill of materials

3. Schematic

The schematic follows the OpenLog Artemis reference and has these parts: Artemis core with crystal and decoupling, ICM-20948 IMU, microSD interface with power switch and level shifting, USB-C with CH340E, LiPo charger and LDOs, Qwiic port, and the debug header. Net labels connect all parts, so one name always means one net.

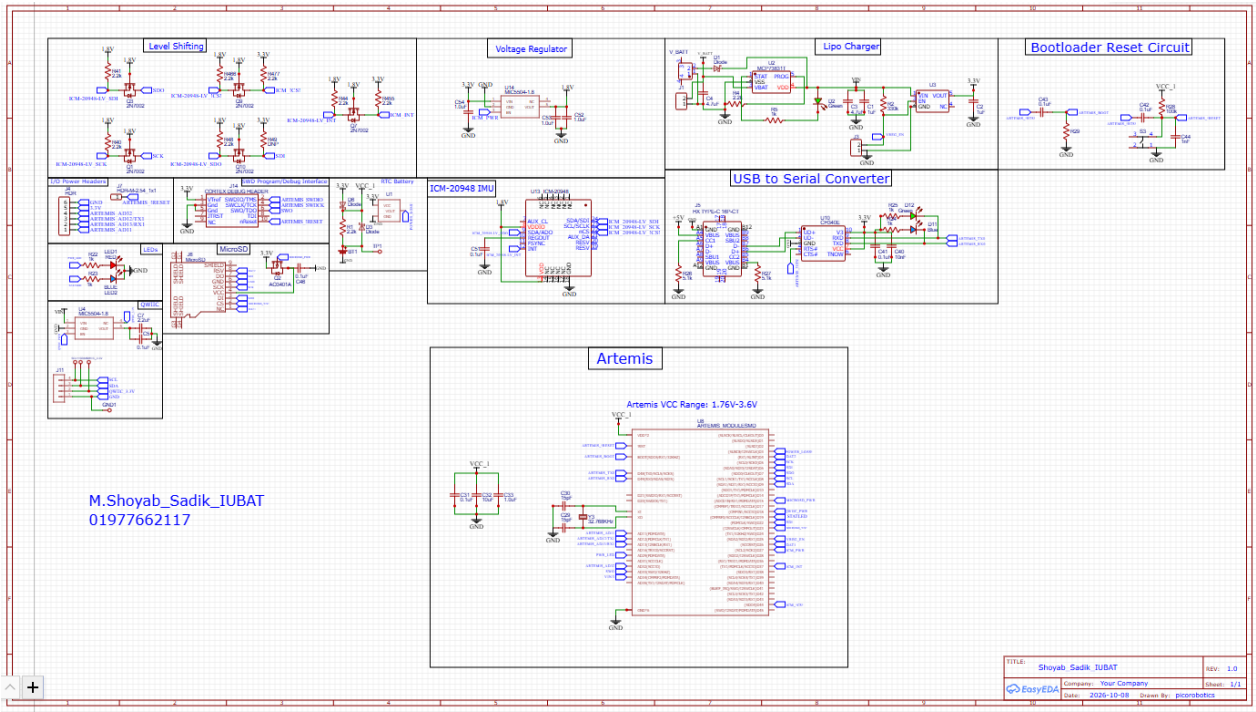


Fig1: Schematic of PCB

4. PCB Design Decisions

The board is [width: 47 x height: 47.117] mm with 2 layers. The Artemis module sits near the board centre with its decoupling capacitors touching its supply pins. The IMU is placed on the opposite side from USB and the charger.

Requirement	**How it was met**
Artemis and decoupling together	[C-caps placed within X mm of U8 supply pins]
IMU away from noisy power and USB	[U13 at X mm from J5, U2 and U3]
microSD near the board edge	[J6 position, slot opening faces the edge]
Short, organised SPI to microSD	[trace length ~X mm, grouped in one bus]
USB connector and protection at edge	[J5 on edge, U10 close to J5]
D+ / D- short, direct, matched	[length difference X mm]
Dedicated power section	[J2, U2, D1/D3, U3, U4 grouped in one corner]
Charging path away from sensor signals	[power section is on the far side from U13]
Qwiic on perimeter, short SDA/SCL	[J11 position, SDA/SCL length X mm]
Programming and serial access	[J14, J1, J3 positions and labels]
Ground plane under MCU, sensor, digital	[GND pour on both layers, stitching vias]
Decoupling close to IC supply pins	[caps within X mm of each IC]
Silkscreen: names, polarity, pin functions	[labels added: J2 +/-, USB, QWIIC, SD, pin names]
Test points, mounting holes, labels	[test points GND, 3.3V, SCL, SDA, TP1; holes added]

Table 2: Requirements checklist

5. Design Rules

Signal traces are 0.25 mm, clearance 0.3 mm. USB D+/D- are routed as a short pair. A GND copper pour is used on both layers with stitching vias.

Rule	Value
Signal trace width	0.25 mm
Power trace width (VBUS, VBAT, 3.3V)	0.25 mm
Clearance	0.30 mm
Via (hole / diameter)	0.35 mm
USB D+/D- length mismatch	< 0.5 mm
Copper pour	GND, top and bottom

Table 3: Design rules used

6. PCB Layout Views

Figure 2 shows the PCB design view and Figures 3, 4 and 5 show the 2D top layer, the 2D bottom layer and the 3D view respectively.

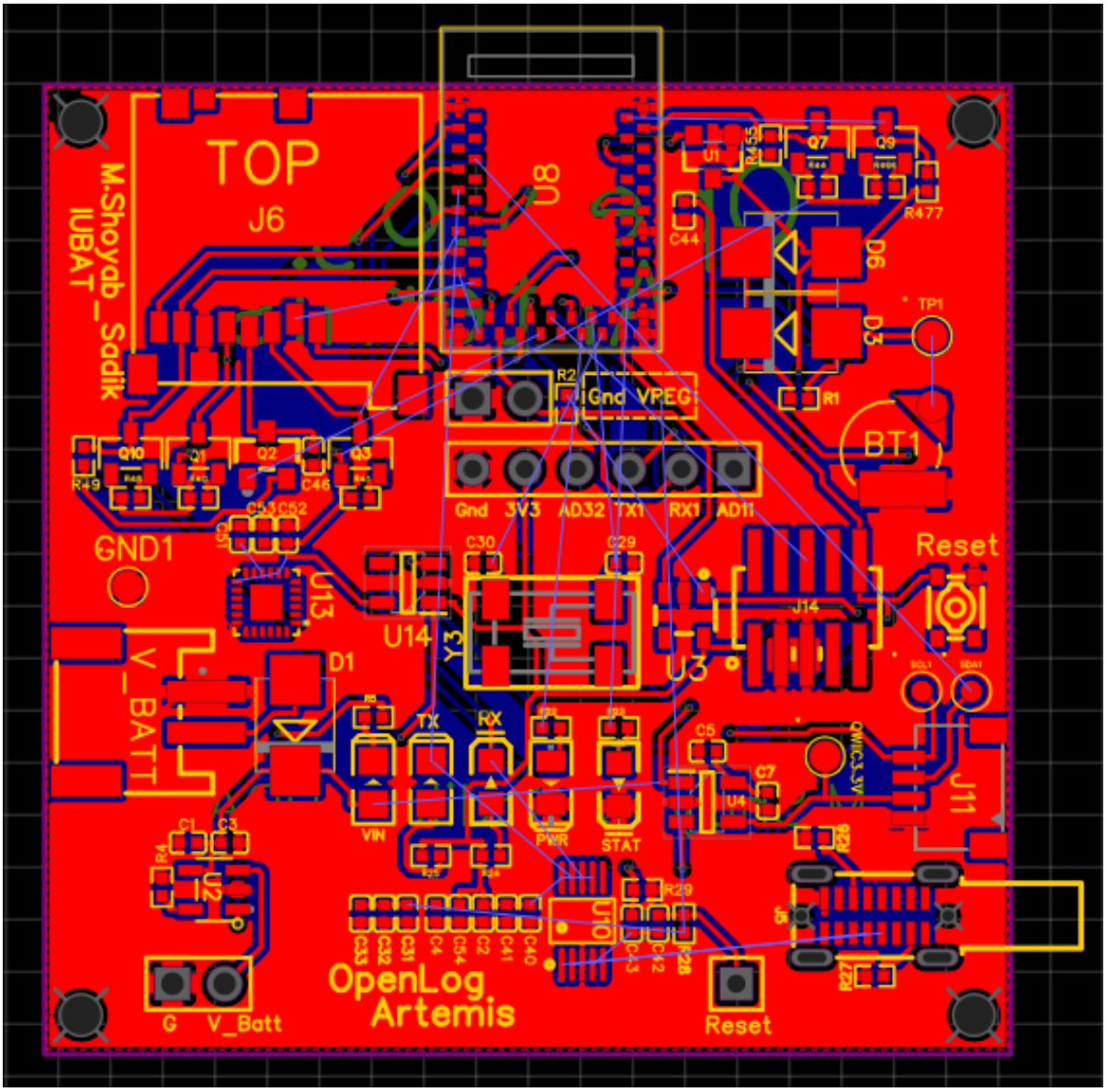


Fig2: PCB Design ViewFig3: Bottom View of PCB

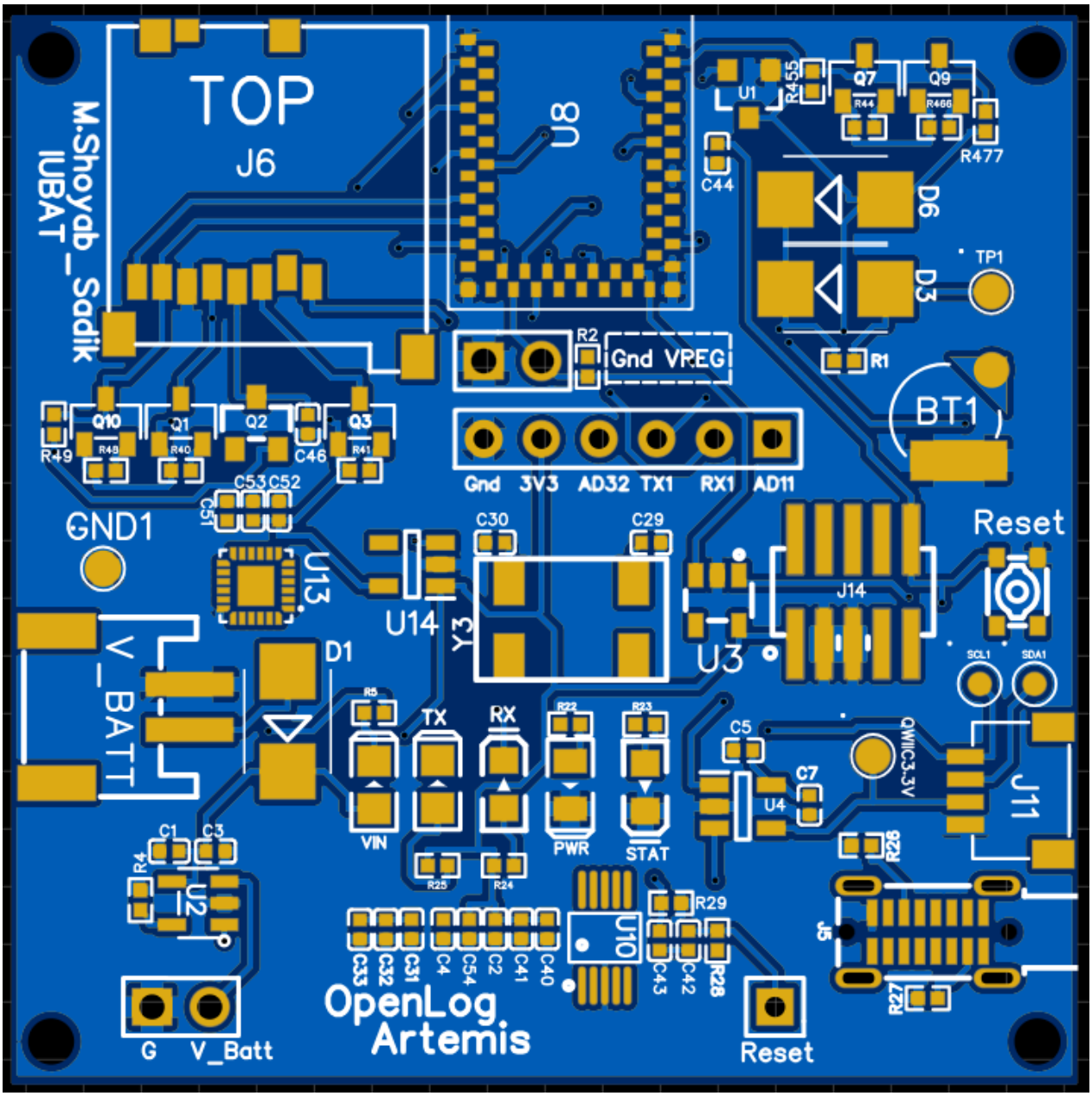


Fig3: Top View of PCB

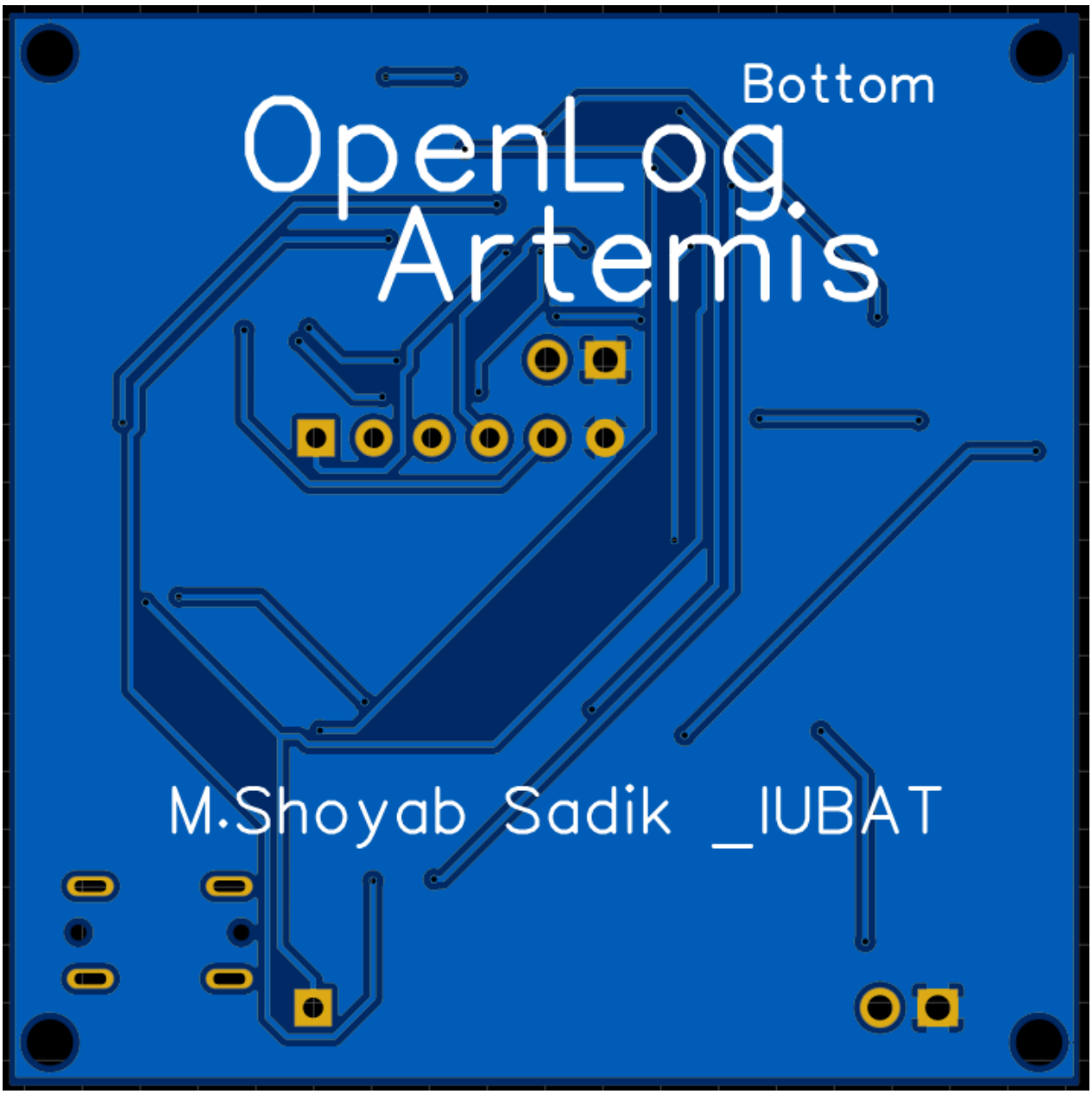


Fig4: Bottom View of PCB

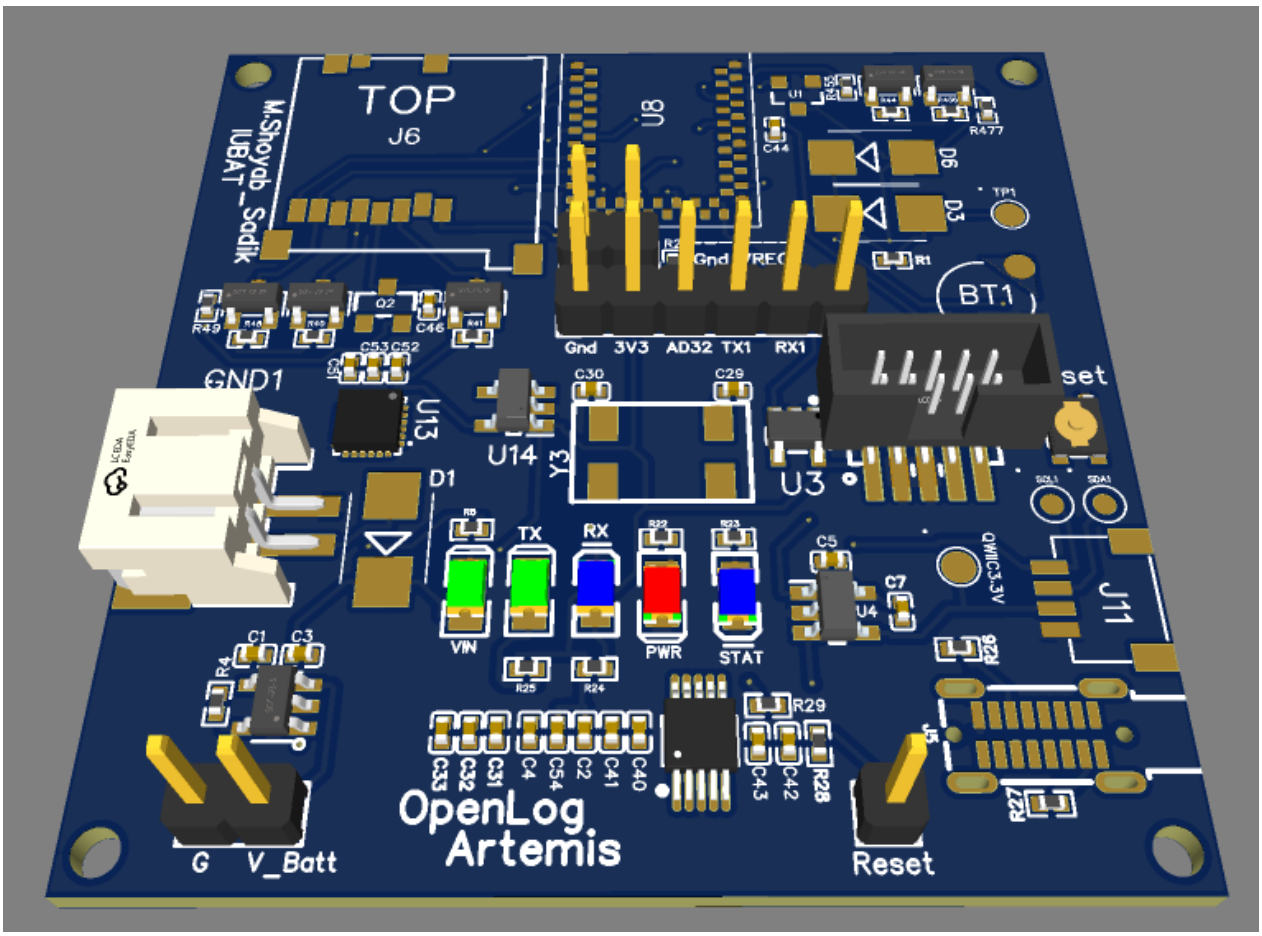


Fig5: 3D View of PCB

7. Conclusion

The design meets the stated requirements: Artemis with close decoupling, IMU away from noise, microSD and USB at the edge, a dedicated power section, short SDA/SCL, accessible debug headers, solid ground coverage, clear silkscreen and a clean DRC.

8. Files Submitted

- ShoyabSadik_IUBAT schematic file
- ShoyabSadik_IUBAT board file
- ShoyabSadik_IUBAT pdf (all layers, one per page)
- ShoyabSadik_IUBAT report (this document)

9. Source Files

GitHub Link: https://github.com/shoyabsadik/iTest-Bangladesh-PCB-Design-Contest-2026